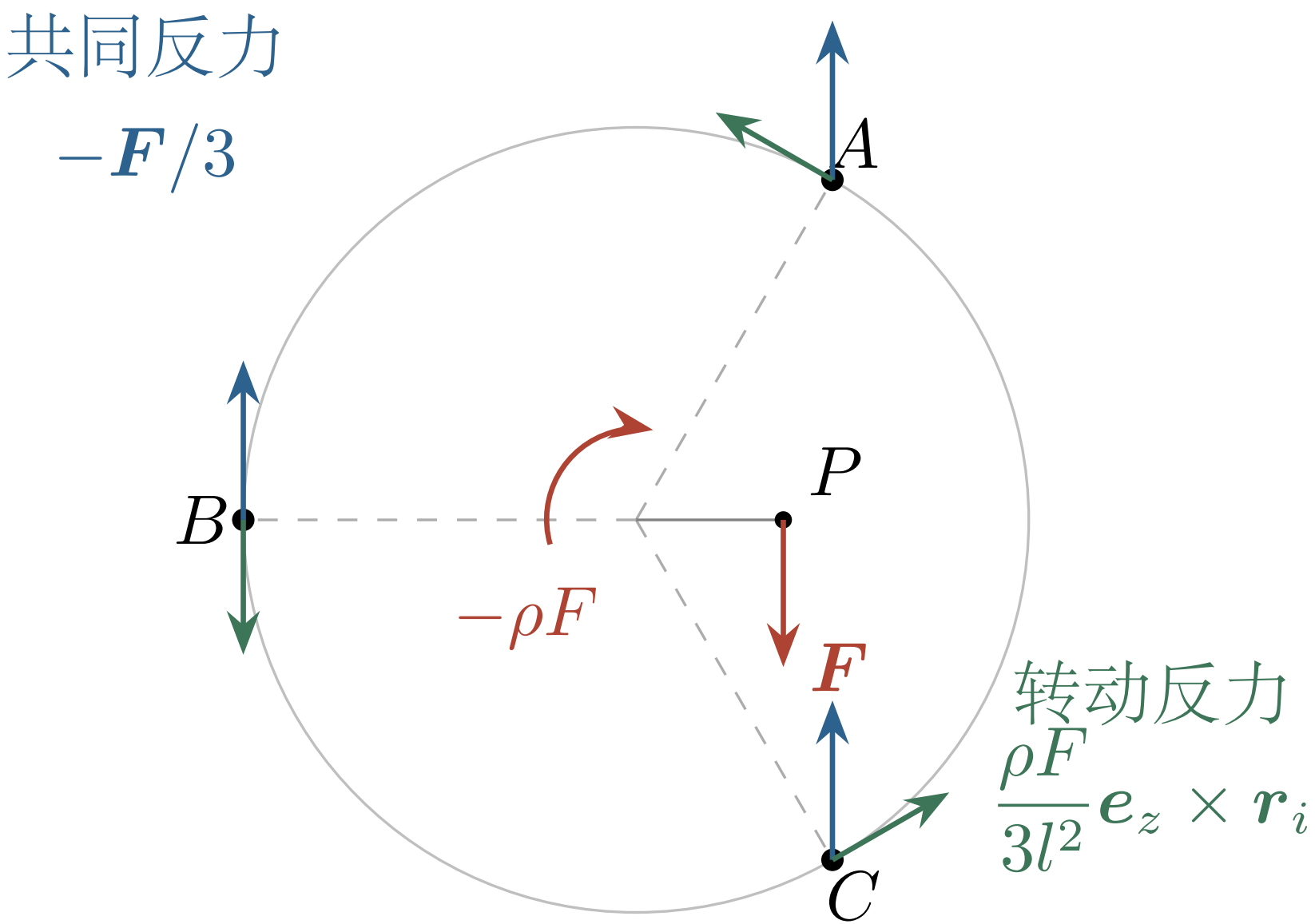

$$\sum_i \boldsymbol{f}_i = -\boldsymbol{F}$$

$$\sum_i (\boldsymbol{r}_i \times \boldsymbol{f}_i)_z = \rho F$$